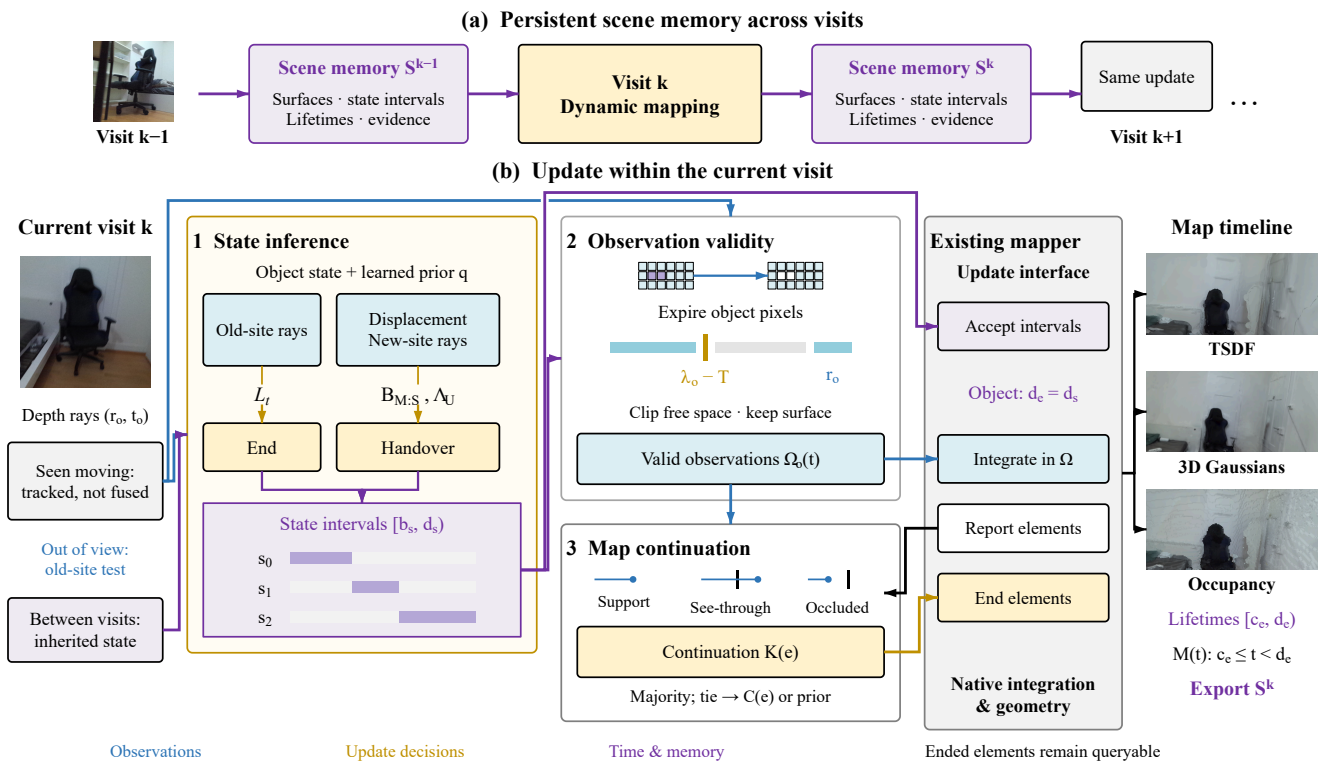

(a) Persistent scene memory across visits
Visit k−1
Scene memory S^{k−1}
Surfaces · state intervals
Lifetimes · evidence
Visit k
Dynamic mapping
Scene memory S^k
Surfaces · state intervals
Lifetimes · evidence
Same update
Visit k+1
(b) Update within the current visit
Current visit k
Depth rays (r_o, t_o)
Seen moving: tracked, not fused
Out of view: old-site test
Between visits: inherited state
1 State inference
Object state + learned prior q
Old-site rays
Displacement New-site rays
L_t
B_{M:S} , Λ_U
End
Handover
State intervals [b_s, d_s)
s0
s1
s2
2 Observation validity
Expire object pixels
λ_o − T
r_o
Clip free space · keep surface
Valid observations Ω_o(t)
3 Map continuation
Support
See-through
Occluded
Continuation K(e)
Majority; tie → C(e) or prior
Existing mapper
Update interface
Accept intervals
Object: d_e = d_s
Integrate in Ω
Report elements
End elements
Native integration & geometry
Map timeline
TSDF
3D Gaussians
Occupancy
Lifetimes [c_e, d_e)
M(t): c_e ≤ t < d_e
Export S^k
Observations
Update decisions
Time & memory
Ended elements remain queryable